INDIAN INSTITUTE OF MANAGEMENT RANCHI

MBA

2026-27

Core Course

COURSE NAME: Corporate Finance

TERM: II

CREDITS: 3

FACULTY DETAILS

Name	Faculty Office Address	Email
Prof. Abhishek Poddar	SK - 303	abhishek.poddar@iimranchi.ac.in
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CONSULTATION TIME FOR STUDENTS: Thursday: 4 pm to 6 pm

COURSE DESCRIPTION:

This is an introductory course for students in the postgraduate program to familiarize them with the financial decisions made by managers/corporations. It broadly covers the following aspects of financial decisions: investment decisions, financing decisions, pay-out decisions, and working capital management. It will help students understand the financial decisions made by a corporation. The course seeks to provide an understanding of the concepts and techniques. Furthermore, it illustrates how these concepts and techniques are applied in practice. The course inputs are what a general manager should know about the finance function.

COURSE OBJECTIVE:

Upon completion of this course, students will be able to develop a critical appreciation and understanding of the financial decisions firms make.

CLO – PLT – ASSESSMENT TOOL MAPPING:

Course Learning Outcome (CLO)	Program Level Trait (PLT)	Assessment Tool(s)
CLO 1: Apply time value of money principles to understand and evaluate the fundamental value of the firm.	T3: Students should be able to learn different methodologies T4: Students should be able to apply appropriate methodologies to solve different business problems	End-term Exam
CLO 2: Estimate the cost of capital and analyze the relationship between risk and return using asset pricing models (CAPM).	T3: Students should be able to learn different methodologies T4: Students should be able to apply appropriate methodologies to solve different business problems	End-term Exam
CLO 3: Evaluate long-term investment decisions and estimate project cash flows utilizing capital budgeting tools (NPV, IRR, MIRR, Payback).	T2: Students should be able to prescribe recommendations to complex business problems T4: Students should be able to apply appropriate methodologies to solve different business problems	End-term Exam
CLO 4: Analyze capital structure theories (MM, Pecking Order) and payout policies (Dividends, Buybacks) to optimize overall firm value.	T1: Students should be able to diagnose complex business problems T2: Students should be able to prescribe recommendations to complex business problems	End-term Exam
CLO 5: Formulate working capital management strategies (CCC) and estimate short-term financing requirements to maintain operational liquidity.	T4: Students should be able to apply appropriate methodologies to solve different business problems T9: Students should be able to identify the interconnectedness across business functions	End-term Exam

PRESCRIBED TEXTBOOK: "*Corporate Finance*" by Stephen A. Ross, Randolph W. Westerfield, Jeffrey Jaffe, Bradford D. Jordan, Ram Kumar Kakani: **McGraw-Hill: 13th edition**

EVALUATION SCHEME:

Evaluation Components as per Program Manual Rules	Detail	Weightage (%)
Quiz	Individual	15%
Attendance	Individual	10%
Group Project	Group	10%
Working with AI (WAI)	Individual	25%
End-term Exam	Individual	40%
Total		**100%**

Group Project: Grading Rubric

Assessment Criteria	Excellent (Full Marks)	Proficient (Partial Marks)	Needs Improvement (Low Marks)	Max Marks
1. Data Collection & Matrix Setup (Excel)	**(2.0 Marks)** Flawless daily data extraction for all 30 equities across 10 sectors (2018–2025). Sheet 1 and 30x30 Covariance Matrix (Sheet 2) are perfectly calculated and labeled.	**(1.0 - 1.5 Marks)** Mostly accurate data collection, but contains minor formulation errors in daily returns or covariance matrix generation.	**(0 - 0.5 Marks)** Significant missing data, incorrect stock selection, or flawed covariance matrix computation.	**2**
2. Risk-Return & Solver Optimization (Excel)	**(3.0 Marks)** Masterful use of Solver (Sheet 3). Portfolio risk, return, and Sharpe ratios are perfectly formulated. Constraints are correctly applied and successfully repeated across all financial years.	**(1.5 - 2.5 Marks)** Formulas are generally correct, but Solver constraints were misapplied on one or two years, or minor errors exist in the risk-free rate/excess return setup.	**(0 - 1.0 Marks)** Flawed portfolio variance/return formulas. Solver failed to run properly, or the multi-year repetition was skipped.	**3**
3. Structure & Methodology (Report)	**(1.5 Marks)** Strictly adheres to the 3-4 page limit. Objective, data description, and methodology are articulated with high academic rigor and clarity.	**(0.75 - 1.0 Marks)** Sections are present but lack MBA-level clarity. Explanations of the portfolio construction methodology are somewhat vague.	**(0 - 0.5 Marks)** Disorganized report, missing required sections, or failed to explain the mathematical methodology used.	**1.5**
4. Visuals, Results & Analysis (Report)	**(2.0 Marks)** Highly professional plotting of the annual Sharpe Ratio. Deep, nuanced analysis comparing pre- and post-COVID-19 portfolio performance and optimal weight shifts.	**(1.0 - 1.5 Marks)** Basic charts provided. Analysis states *what* happened but lacks deep investigation into *why* the Sharpe ratio fluctuated.	**(0 - 0.5 Marks)** Missing or confusing charts. Analysis simply repeats data without interpretation.	**2**
5. Managerial Insights & Compliance	**(1.5 Marks)** Exceptional, actionable financial takeaways. Strict compliance with submission rules (folder names, PDF/Excel formats).	**(0.75 - 1.0 Marks)** Generic conclusions. Minor errors in formatting, file naming conventions, or page limits.	**(0 - 0.5 Marks)** Weak or missing insights. Blatant disregard for submission and file naming instructions.	**1.5**
Total				**10**

INDIVIDUAL ASSIGNMENT DETAILS: Yes (WAI)

WAI ASSIGNMENT INFORMATION:

Based on the firm you are analyzing for the assigned project, you are required to complete the following tasks using AI tools. **Make sure that you clearly mention the references**

Context:

Between 2020 and 2026, Indian firms faced multiple policy and macroeconomic shocks. Analyze how *your assigned firm* responded to these shocks in terms of investment, capital structure, payout, and risk management. Measure how the shocks affected your firm's stock performance and systematic risk (beta). Finally, critically evaluate whether the firm created or destroyed shareholder value, and recommend what they should have done differently.

Major Shocks to Consider (2020–2026)

- **Mar–May 2020:** COVID-19 lockdown → demand/supply collapse.
- **May 2020 – Mar 2021:** Pandemic stimulus & RBI repo at 4% → cheap liquidity.
- **May 2021:** Second COVID wave.
- **Feb 2022:** Russia–Ukraine war → commodity/energy price surge.
- **May 2022 – Feb 2023:** RBI repo rate hikes (4% → 6.5%).
- **Jan 2023:** Adani Group stock crash (Hindenburg).
- **Aug–Sep 2025:** U.S. H-1B visa fee hike ($100,000) & tariff escalation → IT/export shocks.
- **US/Israel – Iran War:** Oil Price Shock

Your Tasks

1. Problem Identification (5%)

- Select **2–3 shocks** most relevant for your firm.
- Justify why these shocks matter for the firm's operations, financing, or risk profile.

2. Data Analysis (10%)

a) Stock Price & Return Data

Download daily or weekly stock price data (2020–2026) for your firm and for **NIFTY 500** as the market index.

Plot: Stock price trend (2020–2026).

Return trend vs market returns (before & after key shocks).

b) Beta Estimation

Run regressions: Run the Capital Asset Pricing Model for:

- **Pre-shock window** (6 months before the event).

- **Post-shock window** (6 months after the event).
- Plot beta (before vs after) for at least **two shocks** relevant to your firm.

c) **Financial Policy Analysis**

Examine how the firm responded to these shocks **Investment:** Capex, WACC.

Capital Structure: Debt/equity ratio, interest coverage

Payout: Dividend policy, buybacks, cash reserves.

You can take the following proxies

Capex = Gross Fixed Assets/Total Assets

Interest coverage= EBIT/Interest expense

3. Critical Reflection (5%)

- Where did the firm **create value** (right choices)?
- Where did it **destroy value** (wrong choices)?
- Use both **data (beta, returns, financials)** and **concepts (systematic risk, WACC, payout smoothing, trade-off theory)** to justify.

4. Recommendations (3%)

- If you were CFO in 2020–2026: what would you have done differently?
- For the next 5 years: what strategy should the firm adopt (reduce leverage, invest more, alter payout, hedge risks)?

5. References (2%)

Working with AI - Guidelines

Deliverable

AI-Integrated Problem-Solving Project: Submission Guidelines

- **Main Submission:** A 3 to 5-page PDF document (excluding annexures) that must include the following sections:
- **Problem Statement:** A clear, concise description of the business problem being addressed.
- **Proposed Solution:** A detailed explanation of how AI will be used to solve the problem. This should include the specific AI tools, models, or techniques to be employed.
- **Implementation & Results:** A summary of the working solution, including data used, and a summary of the results or findings.
- **Analysis & Conclusion:** A critical analysis of the solution's effectiveness, its limitations, ethical considerations, and potential business impact.
- **AI Contribution Statement:** A separate section detailing the use of generative AI tools (e.g., ChatGPT, Gemini, etc.) in the project's development. This should specify which tools were

used, for what purpose (e.g., brainstorming, coding assistance, literature review), and how you verified the output.

Annexure: The actual project deliverables, such as code files, datasets, or interactive dashboards, can be submitted in a folder / within the main PDF document.

Note: Also adhere to the latest "**WAI Evaluation Policy**" of IIM Ranchi to prepare the deliverables

Date of Submission via Google Forms Link: 12/11/2026

GROUP ASSIGNMENT DETAILS: Yes

GROUP ASSIGNMENT INFORMATION:

As part of your group project, which carries a weightage of 10 marks, you are required to construct an equity portfolio and evaluate its performance over a multi-year period.

Please read and follow the guidelines carefully.

Project Objective

To construct a diversified portfolio comprising 30 equities—with 3 equities selected from each of 10 industry sectors—and to estimate the annual Sharpe Ratio for the portfolio.

Period of Study: *1 April 2018 – 31 March 2025* (Covering pre and post COVID-19 period)

Project Guidelines and Steps

Step 1: Industry Selection

Consider the following 10 Nifty industry sectors for portfolio construction:

1. Nifty Auto Index
2. Nifty Bank Index
3. Nifty Chemicals
4. Nifty FMCG Index
5. Nifty Healthcare Index
6. Nifty IT Index
7. Nifty Metal Index
8. Nifty Realty Index
9. Nifty Consumer Durables Index
10. Nifty Oil and Gas Index

Reference: https://www.nseindia.com/products-services/indices-sectoral

Step 2: Equity Selection

From each of the above sectoral indices, select 3 equities from the index constituents.

Example: https://www.niftyindices.com/indices/equity/sectoral-indices/nifty-auto

Step 3: Data Collection

Download the historical daily closing prices of each selected equity.

Example: https://www.nseindia.com/get-quotes/equity?symbol=TATAMOTORS

Step 4: Data Preparation (Sheet 1)

- ✓ Use the “Close Price” of each equity to compute daily returns.
- ✓ Estimate the Average Closing Price and Average Return for each equity.

Step 5: Covariance Analysis (Sheet 2)

- ✓ Using Excel’s *Data Analysis Toolpak*, compute the Covariance of Returns for all equities.
- ✓ Construct the complete (30 × 30) Covariance Matrix and label it as “Covariance_Matrix.”

Step 6: Portfolio Assumptions (Sheet 3)

Use the following assumptions for portfolio calculations:

- ✓ Risk-Free Rate (annual): 6%
- ✓ Number of Trading Days per Year: 252

Transpose the Average Return (Ri) values from Sheet 1 and label the resulting matrix as “Average_Ri.”

Step 7: Risk-Return Estimation (Sheet 3)

- ✓ Compute the daily Risk-Free Rate and calculate Excess Returns (Ri – Rf).
- ✓ Assign equal weights to all equities:

$$\text{Weight per equity} = \frac{1}{30}$$

Label this as the “Weight” matrix.

- ✓ Estimate the Portfolio Return and Portfolio Risk.
- ✓ Compute the Sharpe Ratio using the formula:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Step 8: Optimization Using Solver (Sheet 3)

Use Excel’s Solver tool to maximize the Sharpe Ratio by adjusting the portfolio weights, subject to the following constraints:

- ✓ The sum of all weights = 1
- ✓ Each individual weight > 0

Step 9: Repetition of Analysis

Repeat all the above steps for each financial year within the study period (2018–2025).

Step 10: Plot the estimated Sharpe Ratio w.r.t. each year

Instructions for Project Report Submission

Each group is required to submit the following components as part of the project assessment:

1. **Excel Workbook**

The workbook should contain all relevant calculations and results, organized as follows:

- ✓ **Sheet 1 – Returns:** Daily closing prices, calculated daily returns, average closing prices, and average returns of all selected equities.
- ✓ **Sheet 2 – Covariance Matrix:** Covariance of returns for all 30 equities, presented as a complete (30 × 30) matrix labeled *“Covariance_Matrix.”*

- ✓ **Sheet 3 – Portfolio Analysis:** Calculations for portfolio return, risk, Sharpe ratio, and optimization using Solver, including all assumptions and constraints applied.

2. **Project Report (3–4 Pages)**

A concise report summarizing the following:

- ✓ **Title and Group member** names along with ID
- ✓ **Objective and Scope** of the study.
- ✓ **Data Description** (sources, sample period, and selected sectors/equities).
- ✓ **Methodology** (steps undertaken for portfolio construction, return computation, and Sharpe ratio estimation).
- ✓ **Results and Analysis** (annual Sharpe ratios, optimized portfolio performance, and key observations).
- ✓ **Conclusion** and managerial/financial insights derived from the analysis.

3. **Submission Format**

- The Excel workbook and report should be submitted as two separate folders named as follows:
  - ✓ StudentName_PortfolioProject.xlsx
  - ✓ GroupNumber_PortfolioReport.pdf

Submission Information:

Date of Submission via Google Forms Link: 13/11/2026

SOFTWARE(S) USED (Yes/No) (If yes, please provide details)**: Yes, Excel**

FIELD TRIPS / WORKSHOPS PLANNED (Yes/No) (If yes, please provide details)**:** No

ACADEMIC MISCONDUCT: Please refer to the latest program manual (MBA/MBA-BA/MBA-HRM/IPM/MBA-Executive) provided to you.

ELEMENTS OF SOCIAL IMPACT IN COURSE CURRICULUM (YES/NO) (If yes, please mention session details/case details/other reading material details that have implications to social impact)**:**

END-TERM EXAM FORMAT:

The end-term exam will be pen-and-paper-based. No formula sheets will be provided. You will be allowed to use a calculator.

Date of Quiz Exam (MCQ-based) : 14/11/2026

DETAILED SESSION PLAN:

Session No.	Business Problem	Topic(s) & Discussion Question	Readings (Text Book, Articles, Cases)	Pre-Read
Module 1:	Understanding the concept of value of the firm			
1	What is the Value of the firm, and why should we care about the value of the firm?	Discussion Question: What is the role of Corporate Finance in creating the firm's value?	Textbook: Chapters 1 & 2	
2	How can flawed corporate finance decisions such as sub-optimal capital allocation, excessive leverage, and misaligned payout strategies systematically erode a firm's competitive advantage and destroy its long-term intrinsic value	Discussion Question: What is the role of Corporate Finance in destroying the firm's value?	Textbook: Chapters 1 & 2	
Module 2:	Basic mathematical tools for financial management			
3	What is the time value of money?	Topics: Time Value of Money, Annuities and Perpetuities Hands-on exercise for understanding the Time Value of Money	Textbook: Chapters 3 & 4	Reading: Time Value of Money (HBSP 8299)
Module 3:	Cost of Capital and tools and Techniques for estimating cost of capital			
4	Why are financial markets crucial for firm financing?	Market Structures: Equity Market and Stock markets Why Market structure and regulations are important?	Textbook: Chapters 6 & 7	
5	What are Risk and Return, and why are they important for a firm's cost of equity?	What is the idea of risk and return How to estimate risk of a listed firm?	Textbook: Chapters 11 & 12	
6	How can financial managers evaluate the standalone risk and return of individual investments to strategically	Single Asset Risk and Return; Multiple Asset Risk and Return	Textbook: Chapters 11 & 12	

	combine them into a diversified portfolio that optimizes overall performance and minimizes collective risk?…			
7	How can financial managers accurately estimate a firm's cost of equity by quantifying an asset's sensitivity to systematic market risk to determine the minimum required rate of return for investors?	Introduction to CAPM Estimation and Intuitions behind CAPM Discussion on the application of CAPM	Textbook: Chapters 11 & 12 Case: Beta Management Co. HBSP: 292122	Reading: Applying the Capital Asset Pricing Model HBSP: UV0402 Case: Beta Management Co. HBSP: 292122
8	How do firms estimate the cost of capital?	Levered and unlevered beta	Textbook: Chapter 13	
9	How can financial managers accurately estimate the cost of capital for unlisted firms that lack observable market data by utilizing peer-group proxies and adjusting for specific private-company risks, such as illiquidity and size?	Cost of capital for unlisted firms	Textbook: Chapter 13 Case: Frozen Food Products: Cost of Capital: HBSP: W12324	Reading: Leveraged Betas and the Cost of Equity: HBSP: 288036 Case: Frozen Food Products: Cost of Capital: HBSP: W12324
Module 4	Investment Decision for firm value			
10	How do investment decisions impact the firm value?	Tools and Techniques of capital budgeting decisions: NPV, IRR, MIRR, Payback period	Textbook: Chapters 8 & 9	
11	How can financial managers systematically evaluate and prioritize competing capital investment projects to determine which opportunities	Tools and Techniques of capital budgeting decisions: NPV, IRR, MIRR, Payback period Hands-on practice sessions	Textbook: Chapters 8 & 9	

	will maximize overall firm value and most efficiently recover their initial costs?			
12	How can financial managers effectively apply capital budgeting tools to complex, real-world investment scenarios to ensure optimal resource allocation and maximize long-term firm value?	Application of investment tools	Instructor’s mini case: Capital Budget Analyst	Instructor’s mini case: Capital Budget Analyst
13	How can financial managers accurately forecast a project's Free Cash Flow to the Firm (FCFF), independent of its financing structure, to determine the true operational cash it will generate for all capital providers?	How to estimate the cash flow of a project: FCFF	Instructor’s mini case: Capital Budget Analyst	Instructor’s mini case: New Expansion Project Analysis
14	"How can financial managers systematically quantify the operational and market uncertainties inherent in a proposed capital investment to accurately adjust projected cash flows and determine the appropriate risk-adjusted discount rate?	How to estimate the risk of a project	Instructor’s mini case: New Expansion Project Analysis	Instructor’s mini case: New Expansion Project Analysis
Module 5:	Capital Structure decision for firm value			
15	How does the capital structure of a firm impact its value?	Debt in capital structure, theories of capital structure MM1 & 2	Textbook Chapters 15 and 16	
16	How can financial managers balance the pursuit of a	Optimal capital Structure, Pecking Order theory	Textbook Chapters 15 and 16	

	value-maximizing optimal capital structure with the practical reality of the pecking order theory to determine the most cost-effective sequence for funding operations?			
Module 6:	Payout decisions for firm value and to understand the impact on value for multiple decisions			
17	How does the capital structure of a firm impact its value?	Basics of Dividend& Buyback, Payout Theories	Textbook Chapters: 17	
18	How can financial managers strategically balance the complex interplay between capital structure choices and payout decisions to optimize the firm's funding mix, maintain financial flexibility, and maximize overall shareholder value?	Capital structure and payout decisions are at play	Case: Blaine Kitchenware, Inc.: Capital Structure (HBSP: 4040)	Case: Blaine Kitchenware, Inc.: Capital Structure (HBSP: 4040)
Module 7:	Working capital management (WCM) and its impact on firm value			
19	Does WCM impact firm value?	Short-Term Financing CCC	Textbook Chapters: 19	Case: Dell's Working Capital (HBSP: 201029)
20	How can financial managers accurately forecast working capital requirements to determine the exact level of short-term financing needed to sustain daily operations and maintain optimal liquidity?	Estimation of short-term financing needs	Textbook Chapters: 19	Data Case (page no. 637 of the Text Book)